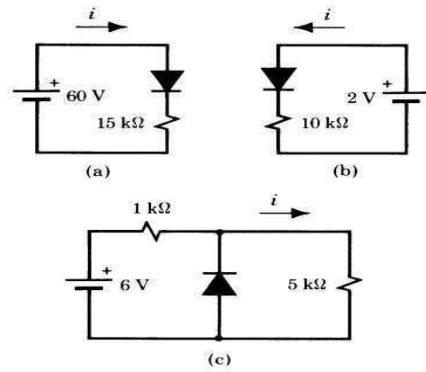


ELL 101: Introduction to Electrical Engineering

- 1) Find the current i for each of the following circuits of germanium diodes ($v_T = 0.7$ V).



- 2) For the circuit shown in Fig. 2(a), sketch the waveforms for v_L , v_D for the given source voltage v_S as shown in Fig. 2(b).

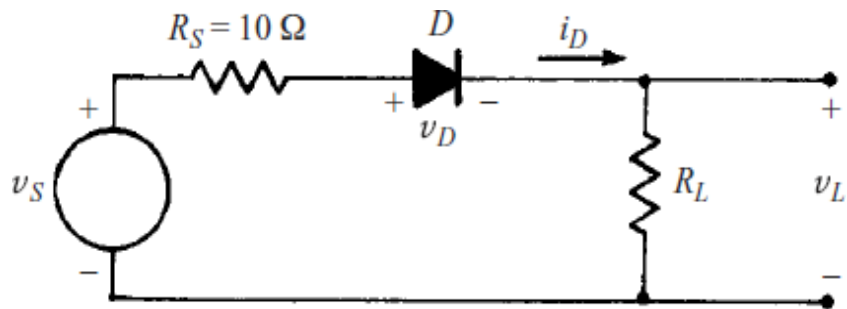


Fig. 2(a)

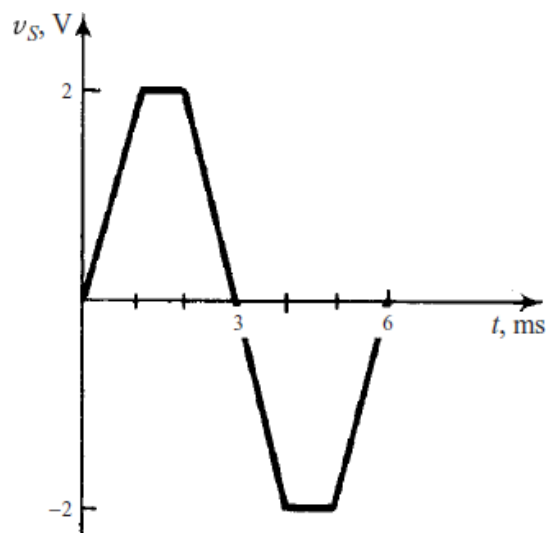
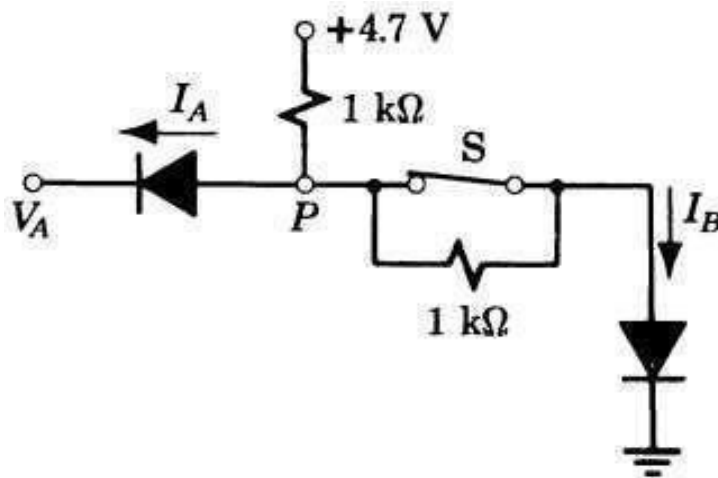


Fig. 2(b)

- 3) The switching circuit of the following figure contains two silicon ($v_t = 0.3\text{ V}$) Diodes.
- For switch S closed, estimate I_A and I_B for $V_A = -1, -0.1, +1$ and $+2\text{ V}$.
 - For switch S open, estimate I_A and I_B for $V_A = +1\text{ V}$.



- 4) For the circuit shown in Fig. 4(a), v_i is shown in Fig. 4(b). Let $V_b = 10\text{ V}$ and $R = 1\text{ k}\Omega$.
- Sketch v_o vs t .
 - Sketch v_o v/s v_i on the sample plot (transfer characteristic)

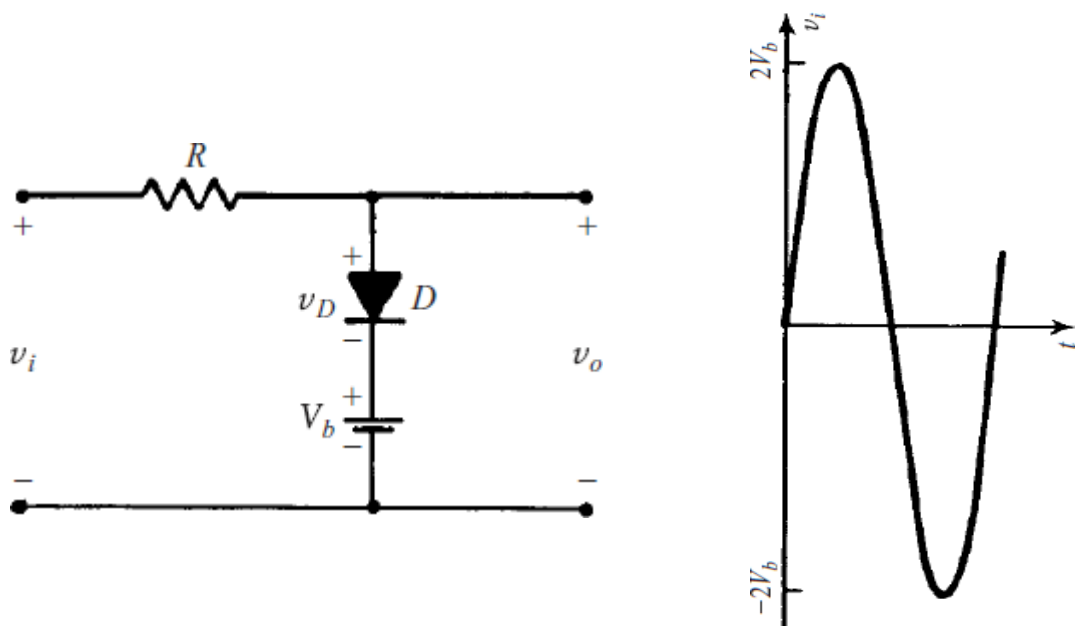


Fig. 4 (a), (b).

- 5) For the circuit shown in Fig. 5(a), a triangular input waveform is given as input (Fig. 5(b)). If capacitor C is initially uncharged, and $V_B = 0$, Sketch the output waveform V_o vs t .

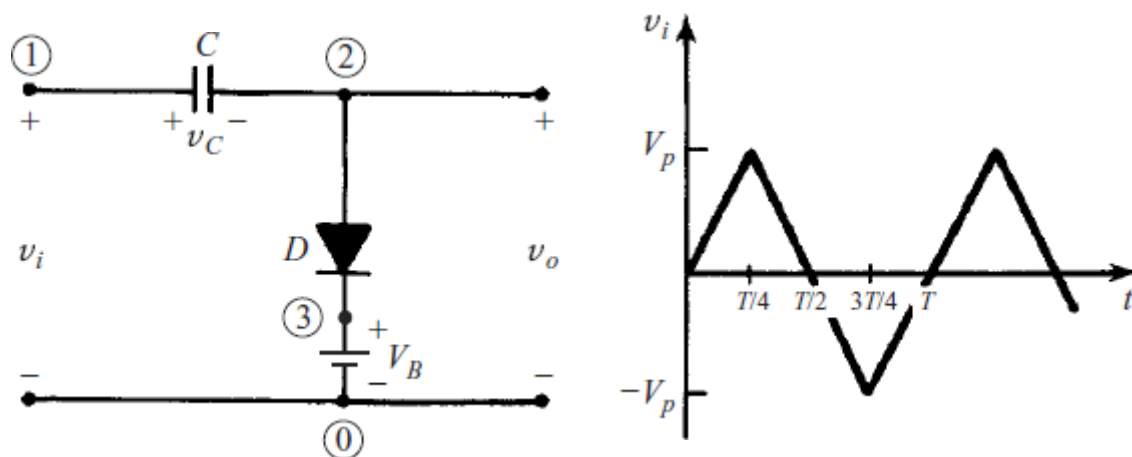


Fig. 5(a), (b)

- 6) *For the circuit shown in Fig. 6(a), v_S is a 10V square wave of period 4ms (shown in Fig. 6(b)), $R = 100$ ohms, $C = 20 \mu F$. Determine $v_C(t)$ for the for each time period, if capacitor is initially uncharged and diode is ideal.

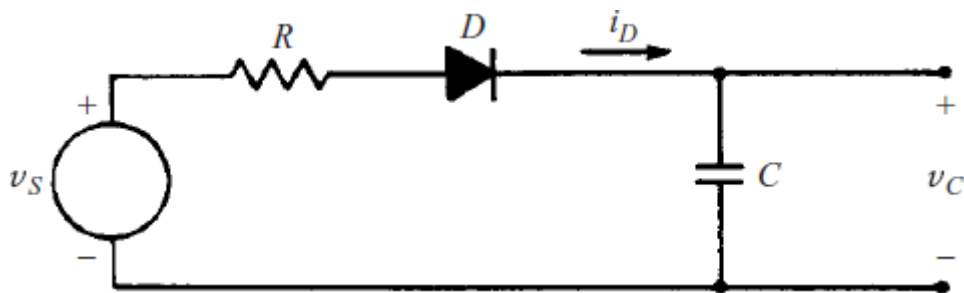


Fig. 6(a).

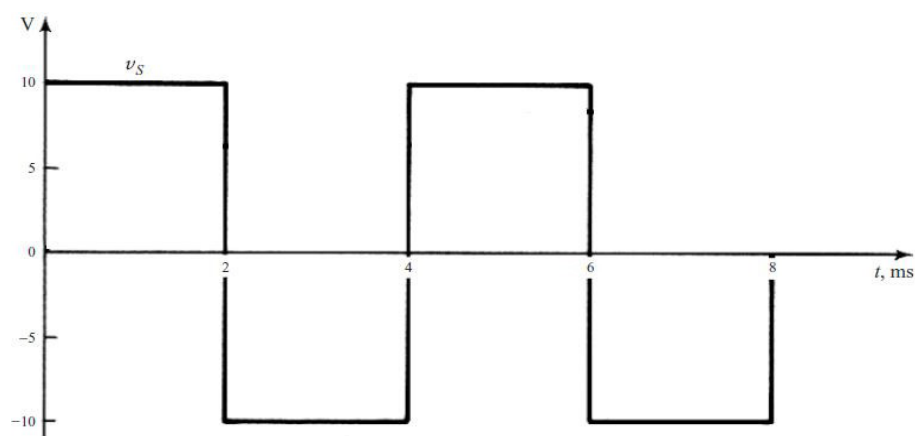


Fig. 6(b)

7) The diodes in Fig. 7. are ideal. Compute V_2 if $-20 \leq V_1 \leq 20V$.

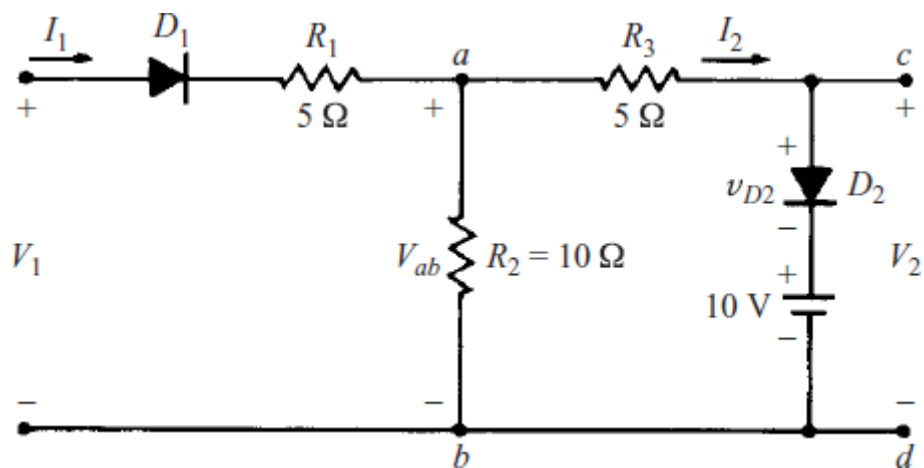


Fig. 7

Q8. *The zener diode in the circuit has a constant reverse breakdown voltage, $V_Z = 8.2 V$, $75 mA \leq i_Z \leq 1 A$. If $R_L = 9 ohms$, $R_S = ?$ so that v_L is regulated at $8.2 V$. Given that nominal value of $V_b = 12 V$ but it varies by $\pm 10\%$ of its nominal value.

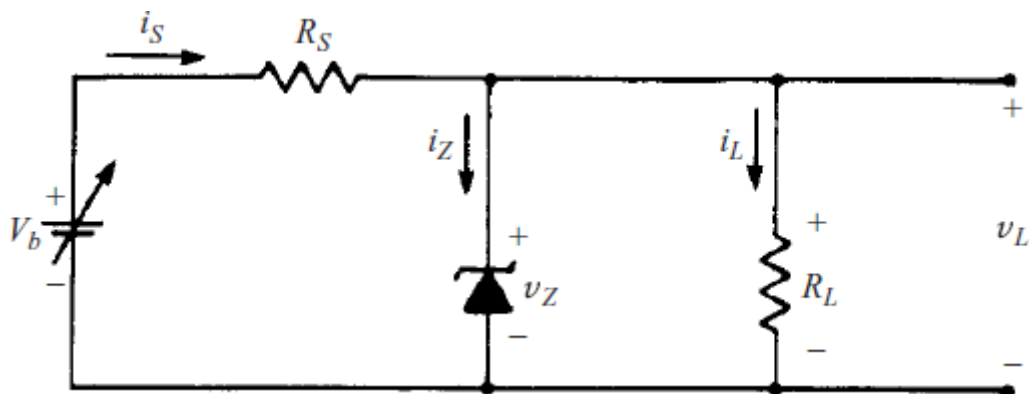


Fig. 8